A MICRO PROJECT REPORT ON

AUTOMATIC RAILWAY LEVEL CROSSING OPENING AND CLOSING SYSTEM USING STM32 MICROCONTROLLER

Submitted for partial fulfillment of the requirements for the subject PBCST504

MICROCONTROLLERS

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IN

COMPUTER SCIENCE AND ENGINEERING

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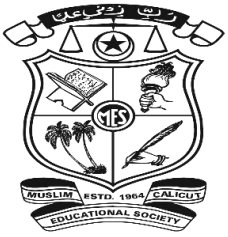
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CERTIFICATE

This is to certify that the project report entitled **"**Automatic railway level crossing opening and closing system using stm32 microcontroller**"**, submitted by Adithyan (MES24CS011) , Ayisha Minha (MES24CS029) , Fathima Jebin (MES24CS046) , Fathima Mehrin (MES24CS049) , Pranav (MES24CS0) to APJ Abdul Kalam Technological University, Kerala, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering, is a bonafide record of the project work carried out by them under my guidance and supervision.

The work presented in this report has been completed as part of the academic requirements of the Microcontrollers course.

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DECLARATION

We, the undersigned, hereby declare that the project report **"**Automatic railway level crossing opening and closing system using stm32 microcontroller**"**, submitted for partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of APJ Abdul Kalam Technological University, Kerala, is a bonafide work carried out by us under the supervision of Fathima Sanam U V , Department of Computer Science and Engineering, MES College of Engineering, Kuttippuram.

This submission represents our work in our own words, and wherever the ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to the principles of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact or source in this submission.

We understand that any violation of the above will be a cause for disciplinary action by the institution and/or the University. This report has not previously formed the basis for the award of any degree, diploma or similar qualification from any other University.

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ABSTRACT

Railway level crossings are locations where railway tracks intersect roads used by vehicles and pedestrians. The safe operation of these crossings requires proper coordination between train movement and road traffic. Conventional manually operated railway gates depend on human intervention to detect approaching trains, activate warnings and operate the barriers. Delays in operation and human errors may increase the risk of accidents at railway crossings. Therefore, automation can help demonstrate a more systematic approach to controlling railway gates.

The **Automatic Railway Level Crossing Opening and Closing System Using STM32 Microcontroller** is designed to automate the basic operation of a railway crossing using an embedded control system. The proposed system uses an STM32 Blue Pill development board based on the STM32F103C8T6 microcontroller as its main processing unit. Two Infrared (IR) sensors are used to detect the approaching train and identify when it has passed the crossing. A servo motor is used to operate the miniature railway gate, while a buzzer and red and green LEDs provide audible and visual warnings.

When the approaching train is detected by the first IR sensor, the STM32 microcontroller processes the input and activates the warning system. The red LED and buzzer are switched on, and the servo motor moves the gate to the closed position. The gate remains closed while the train passes through the crossing. After the train reaches the second IR sensor and the clearance condition is detected, the controller stops the warning, opens the gate and activates the green LED.

The project demonstrates the application of microcontroller-based automation in railway crossing management. It provides practical understanding of sensor interfacing, GPIO control, servo motor operation and embedded C programming using the STM32 platform. The system is initially studied through simulation, followed by physical hardware implementation and testing. This prototype is intended for educational demonstration and is not designed for direct use in real railway infrastructure.

The implementation of this system also provides an opportunity to understand the integration of hardware components with embedded software. By combining sensor inputs, programmed decision-making and output devices, the project demonstrates how an automated control mechanism can reduce dependence on manual operation. The prototype highlights the importance of accurate train detection and proper sequencing of gate operations in improving the safety and efficiency of railway level crossing management.

CONTEXT

Chapter 1: INTRODUCTION 1
1.1 Project Overview
1.2 Problem Statement
1.3 Objectives
1.4 Scope of the Project
Chapter 2: LITERATURE REVIEW 3
2.1 Railway Level Crossing Operation
2.2 Manual Railway Gate Operation
2.3 Sensor-Based Train Detection
2.4 Microcontroller-Based Automation
Chapter 3: SYSTEM REQUIREMENTS 5
3.1 Hardware Requirements
3.2 Software Requirements
Chapter 4: SYSTEM DESIGN AND ARCHITECTURE 7
4.1 System Architecture
4.2 Working Principle
4.3 System Flow
Chapter 5: IMPLEMENTATION 9
5.1 Hardware Integration
5.2 Software Implementation
5.3 Train Detection and Gate Control
5.4 Output Control and System Operation
Chapter 6: TESTING 11
6.1 Testing Methodology
6.2 Test Cases
6.3 Performance Parameters
Chapter 7: RESULTS AND DISCUSSION 13
7.1 Functional Testing Results
7.2 Results
7.3 Analysis
7.4 Key Findings
Chapter 8: CONCLUSION AND FUTURE SCOPE 15
8.1 Conclusion
8.2 Future Scope
APPENDICES
Appendix A: Component Summary 16
Appendix B: System Flow & Architecture 17
REFERENCES 19

Chapter 1: INTRODUCTION

1.1 Project Overview

Railway transportation plays an important role in the movement of passengers and goods. Railway tracks frequently intersect roads at level crossings, where road vehicles and pedestrians must share the crossing area with trains. Since trains require a considerable stopping distance and cannot stop immediately, it is essential to control road traffic whenever a train approaches a crossing.

Traditional railway level crossings may depend on manual gate operation, where a railway operator is responsible for monitoring train movement and controlling the gate. The operator must ensure that the gate is closed before the train reaches the crossing and remains closed until the train has completely passed. Any delay or mistake during this process may increase the possibility of accidents.

The Automatic Railway Level Crossing Opening and Closing System Using STM32 Microcontroller is proposed as an automated solution to demonstrate the basic operation of a railway crossing. The system uses an STM32 Blue Pill board based on the STM32F103C8T6 microcontroller as the central processing unit. Two Infrared (IR) sensors are used to detect the approaching train and its clearance after passing the crossing.

The first IR sensor is positioned before the railway crossing to detect the arrival of the train. When the train is detected, the microcontroller activates a buzzer and red LED to provide warning indications. At the same time, a servo motor is controlled to lower the miniature railway gate and prevent road traffic from crossing.

A second IR sensor is positioned beyond the crossing to detect when the train has passed the crossing area. Once the clearance condition is detected, the STM32 microcontroller commands the servo motor to raise the gate. The buzzer is switched off, the red LED is turned off and the green LED is activated to indicate that the crossing is open.

The proposed system demonstrates how sensors, actuators and a microcontroller can work together to automate a railway level crossing. It provides practical knowledge of embedded systems, sensor interfacing, GPIO operations and servo motor control using the STM32 platform.

1.2 Problem Statement

Railway level crossings require proper coordination between train movement and road traffic to ensure safety. In conventional manually operated systems, the railway gate must be closed at the appropriate time when a train approaches and reopened only after the train has passed the crossing.

Manual operation involves continuous monitoring and timely human intervention. Delayed gate closing, incorrect timing and human errors may increase the risk of accidents. In

addition, the gate must remain closed until the entire train has cleared the crossing. Reopening the gate too early may create a dangerous situation for road users.

Therefore, there is a need to demonstrate an automated railway gate control system that can detect train movement and operate the gate according to predefined conditions without requiring continuous manual operation.

This project addresses the problem by developing a sensor-based railway crossing model using an STM32 microcontroller. The system uses IR sensors to detect train approach and clearance, a servo motor to control gate movement, and a buzzer with LED indicators to provide warning signals.

1.3 Objectives

The primary objective of this project is to design and develop an automatic railway level crossing model using an STM32 microcontroller.

The specific objectives are:

- To design an automated railway gate control system using the STM32 Blue Pill development board.
- To detect the approaching train using an IR sensor.
- To identify when the train has passed the crossing using a second IR sensor.
- To automatically close the railway gate using a servo motor when a train is detected.
- To activate a buzzer and red LED to provide warning indications during train movement.
- To automatically reopen the gate after the train has cleared the crossing.
- To use a green LED to indicate the normal open-gate condition.
- To understand the practical application of embedded systems and microcontroller-based automation.

1.4 Scope of the Project

The scope of this project is to develop a small-scale automatic railway level crossing model using an STM32 microcontroller. The system focuses on detecting the approach and clearance of a train and controlling the opening and closing of a miniature railway gate.

The project uses two IR sensors as input devices, an STM32 Blue Pill board as the main controller, and a servo motor as the gate-control mechanism. A buzzer and two LEDs are included to provide audible and visual indications of the crossing status.

The proposed system demonstrates the basic working sequence of an automated railway crossing, starting from train detection to warning activation, gate closing, train passage and gate reopening. The design can be used as an educational model to understand sensor-based automation and embedded-system development.

Chapter 2: LITERATURE REVIEW

2.1 Railway Level Crossing Operation

Railway level crossings are points where railway tracks intersect roads or pathways at the same level. These crossings require proper traffic management to ensure the safe movement of trains as well as road users. Since trains travel at high speeds and require a long distance to stop, it is important to provide sufficient warning before a train reaches a crossing.

A railway level crossing generally consists of a gate or barrier, warning lights, audible alarms and a control mechanism. When a train approaches, the crossing must be secured by closing the gate and activating warning signals. After the train passes completely, the gate can be reopened to allow road traffic to move.

The proper functioning of these components is essential for reducing the possibility of collisions between trains and road vehicles. Automation can improve the coordination of these operations by using sensors and electronic controllers to perform predefined actions.

2.2 Manual Railway Gate Operation

In conventional railway level crossing systems, gate operation may be performed manually by a railway operator. The operator monitors train movement and closes the gate when a train approaches. Warning signals are provided to alert road users, and the gate is reopened after the train has passed.

Although manual operation is a commonly used method, it depends on the attention and timely action of the operator. Delayed operation, human error and improper coordination may increase the risk of accidents. Continuous monitoring is also required to ensure that the gate is operated at the correct time.

An automated railway gate system can reduce the need for continuous manual intervention by using electronic sensors to detect train movement. The controller can process the sensor signals and operate the gate and warning indicators according to predefined conditions.

2.3 Sensor-Based Train Detection

Sensors play an important role in automated railway crossing systems because they provide information about the movement of trains. Different types of sensors can be used to detect the presence or movement of a train and send signals to a control unit.

Infrared (IR) sensors are electronic devices that use infrared light to detect the presence of an object. They generally consist of an infrared transmitter and a receiver. When an object interrupts or reflects the infrared signal, the sensor produces an output that can be read by a microcontroller.

In the proposed system, two IR sensors are used for train detection. The first sensor detects the approaching train and sends a signal to the STM32 microcontroller. Based on this input, the controller activates the warning system and closes the railway gate. The second sensor is used to identify the train's passage beyond the crossing and provide the clearance input required to reopen the gate.

The use of two sensors helps demonstrate the basic sequence of train detection and clearance monitoring in an automated railway crossing model.

2.4 Microcontroller-Based Automation

Microcontrollers are compact programmable devices used to control electronic systems. They contain a processor, memory and input/output peripherals that allow them to receive signals from sensors and control external devices.

In automation applications, a microcontroller receives input signals, processes them according to a programmed logic and generates suitable output signals. This makes it possible to automate operations that would otherwise require manual control.

The proposed railway level crossing system uses the STM32 Blue Pill development board, which is based on the STM32F103C8T6 microcontroller. It acts as the central processing unit and coordinates the operation of the IR sensors, servo motor, buzzer and LEDs.

When the first IR sensor detects a train, the microcontroller processes the input and activates the red LED and buzzer. It also sends a control signal to the servo motor to close the gate. When the second sensor detects the train's clearance, the controller stops the warning indications and operates the servo motor to reopen the gate. The green LED indicates that the crossing is open.

Microcontroller-based automation provides a practical method for integrating sensors and output devices into a single control system. This project demonstrates the application of STM32 technology in a railway crossing model and provides practical experience in embedded programming and hardware interfacing.

Chapter 3: SYSTEM REQUIREMENTS

3.1 Hardware Requirements

The proposed system requires various electronic components to detect train movement, process sensor signals and control the railway gate. Each hardware component performs a specific function in the overall operation of the system.

1. STM32 Blue Pill Board

The STM32 Blue Pill is the main development board used in the project. It is based on the STM32F103C8T6 microcontroller, which uses an ARM Cortex-M3 processor. It receives signals from the IR sensors, processes the inputs according to the programmed logic and controls the connected output devices. It acts as the central processing unit of the railway crossing system.

2. IR Sensors

Infrared sensors are used to detect the presence of the train at different positions. The proposed system uses two IR sensors. The first sensor detects the approaching train and sends a signal to the microcontroller to initiate the gate-closing process. The second sensor detects the train after it passes the crossing and provides the clearance input required to reopen the gate.

3. SG90 Servo Motor

The SG90 servo motor is used to control the movement of the miniature railway gate. It receives control signals from the STM32 microcontroller and rotates to predefined angular positions. When a train is detected, the servo motor moves the gate to the closed position. After the train clears the crossing, it rotates back to raise the gate.

4. Buzzer

A buzzer is used to produce an audible warning when a train approaches the railway crossing. It is activated along with the red LED during the gate-closing and train-passing conditions. This provides an additional warning to indicate that the crossing is closed.

5. Red LED

The red LED indicates that the railway crossing is closed and that road users should wait. It is activated when the first IR sensor detects an approaching train and remains active until the clearance condition is detected.

6. Green LED

The green LED indicates that the railway crossing is open and road traffic can proceed. It is activated after the train has passed the crossing and the gate has been raised.

7. ST-Link V2 Programmer

The ST-Link V2 compatible programmer is used to upload the program to the STM32 Blue Pill board. It provides an interface between the computer and the microcontroller during programming and debugging.

8. Breadboard and Jumper Wires

A breadboard is used to connect the electronic components without permanent soldering. Jumper wires are used to establish electrical connections between the STM32 board, IR sensors, servo motor, buzzer and LEDs.

9. Resistors

Resistors are used with the LEDs to limit the current flowing through them and protect the LEDs from damage.

10. Regulated 5V Power Supply

A regulated 5V power supply is used to provide a suitable power source for components such as the SG90 servo motor. A stable power supply is important for reliable operation. The servo motor's power supply ground and the STM32 ground must be connected together to ensure proper signal reference.

3.2 Software Requirements

The development of the proposed system requires suitable software tools for programming, uploading and testing the STM32 microcontroller.

1. Embedded C Programming Language

Embedded C is used to develop the program that controls the operation of the railway crossing system. The program reads the inputs from the IR sensors and controls the servo motor, buzzer and LED indicators according to the required sequence.

2. STM32 Development Environment

An STM32-compatible development environment is required to write, compile and upload the embedded program to the STM32 Blue Pill board. It helps in configuring the microcontroller pins and developing the control logic.

3. STM32CubeIDE

STM32CubeIDE is an integrated development environment used for STM32 microcontroller development. It provides facilities for writing code, configuring peripherals, compiling programs and debugging applications. It can be used for developing firmware for the STM32F103C8T6 microcontroller.

Chapter 4: SYSTEM DESIGN AND ARCHITECTURE

4.1 System Architecture

The proposed system is designed using an STM32 Blue Pill development board as the central processing unit. The system consists of two IR sensors, an SG90 servo motor, a buzzer, red and green LEDs, and a regulated power supply. Each component is connected to the microcontroller to perform a specific function in the automatic railway gate operation.

The two IR sensors act as input devices and are positioned at suitable locations along the railway track. The first IR sensor detects the approaching train and sends a signal to the STM32 microcontroller. The second IR sensor detects the train after it passes the crossing and provides the clearance input.

The STM32 microcontroller processes the signals received from the sensors according to the programmed instructions. Based on the sensor inputs, it controls the servo motor, buzzer and LED indicators. The SG90 servo motor operates the miniature railway gate by moving it between the open and closed positions.

The buzzer and red LED provide warning indications when the train approaches the crossing. The green LED indicates that the crossing is open after the train has passed. A regulated 5V power supply is used for the servo motor, while the STM32 board provides the control signals required for the connected components.

Thus, the system architecture combines input sensors, a processing unit and output devices to achieve automatic railway gate operation.

Main components of the system:

- **Input section:** Two IR sensors for train detection.
- **Processing section:** STM32 Blue Pill microcontroller.
- **Output section:** SG90 servo motor, buzzer, red LED and green LED.
- **Power section:** Suitable power supply for the microcontroller and connected components.

4.2 Working Principle

The system works based on the signals received from the two IR sensors. The STM32 microcontroller continuously monitors the sensor inputs and controls the connected output devices according to the programmed logic.

Initially, the railway gate remains in the open position. The servo motor maintains the gate in its raised position, the green LED remains ON, and the red LED and buzzer remain OFF. This indicates that road traffic can pass through the crossing.

When a train approaches the railway crossing, the first IR sensor detects its presence and sends a signal to the STM32 microcontroller. The controller processes this signal and activates the warning system. The red LED turns ON and the buzzer produces an audible warning. At the same time, the servo motor rotates to lower the railway gate and close the crossing.

The gate remains closed while the train passes through the crossing. After the train passes the crossing and reaches the second IR sensor, the sensor detects its presence and sends a signal to the microcontroller. The controller processes this input as the train-clearance condition.

Once the clearance condition is confirmed, the microcontroller switches OFF the buzzer and red LED. It then controls the servo motor to raise the railway gate to its original open position. The green LED turns ON to indicate that the crossing is open and road traffic can proceed.

In this way, the system performs the complete sequence of train detection, warning activation, gate closing, train passage and gate opening automatically.

4.3 System Flow

The system follows a predefined sequence of operations to control the railway gate based on the inputs received from the IR sensors.

The sequence of operation is as follows:

1. **System Initialization:** The STM32 microcontroller initializes the connected input and output devices. The gate starts in the open position, and the green LED is activated.
2. **Train Detection:** The first IR sensor continuously monitors for an approaching train. When the train is detected, the sensor sends a signal to the microcontroller.
3. **Warning Activation:** The STM32 processes the sensor input and activates the red LED and buzzer to warn road users.
4. **Gate Closing:** The microcontroller sends a control signal to the SG90 servo motor, which moves the railway gate to the closed position.
5. **Train Passage:** The gate remains closed while the train passes through the crossing.
6. **Clearance Detection:** The second IR sensor detects the train after it passes the crossing and sends the corresponding signal to the microcontroller.
7. **Gate Opening:** After the clearance condition is detected, the microcontroller stops the warning signals and controls the servo motor to raise the gate.
8. **Normal Condition:** The green LED turns ON, indicating that the railway crossing is open and the system returns to its initial condition.

This sequence allows the system to automatically control the miniature railway gate based on train movement and sensor inputs.

Chapter 5: IMPLEMENTATION

5.1 Hardware Integration

The hardware implementation of the automatic railway level crossing system involves connecting the STM32 Blue Pill microcontroller with the required input and output components. The two IR sensors are connected to the input pins of the microcontroller to detect the approaching train and the clearance of the train after passing the crossing.

The SG90 servo motor is connected to the STM32 to control the opening and closing movement of the railway gate. A buzzer is used to provide an audio warning, while red and green LEDs indicate the current status of the crossing. The red LED indicates that the gate is closed, and the green LED indicates that the gate is open.

All the components are connected using a breadboard and jumper wires. A regulated 5V power supply is used for the servo motor, with a common ground connection between the power supply and the STM32 board.

5.2 Software Implementation

The software implementation is developed using Embedded C programming. The program is designed to receive signals from the IR sensors and control the output components according to the detected condition.

Initially, the system is initialized with the gate in the open position. The green LED remains ON, while the red LED and buzzer remain OFF. The microcontroller continuously monitors the signals from the IR sensors.

When the first IR sensor detects an approaching train, the STM32 processes the input and activates the warning system. The buzzer starts producing an alert sound, the red LED turns ON, and the servo motor moves the gate to the closed position.

After the train passes the crossing, the second IR sensor detects the clearance condition. The microcontroller then stops the buzzer, turns OFF the red LED, activates the green LED, and controls the servo motor to raise the gate.

5.3 Train Detection and Gate Control

Train detection is performed using two IR sensors placed at suitable positions along the railway track. The first sensor is responsible for detecting the approaching train, while the second sensor is used to identify when the train has passed the crossing.

The STM32 Blue Pill receives signals from these sensors and makes decisions based on the programmed conditions. When an approaching train is detected, the gate closing sequence is initiated. The gate remains closed during train passage to prevent crossing.

Once the clearance condition is detected, the gate opening sequence begins. This coordinated operation helps reduce the need for manual control and provides an automated method of managing railway level crossing operations.

5.4 Output Control and System Operation

The output devices are controlled by the STM32 microcontroller based on the input received from the IR sensors.

The servo motor controls the movement of the miniature railway gate. The buzzer provides an audio warning when a train approaches the crossing. The red and green LEDs provide visual indications of the gate status.

The complete system operates through a continuous control sequence involving train detection, warning activation, gate closing, train passage, clearance detection, and gate opening. This sequence demonstrates the integration of sensors, a microcontroller, and output devices in an automated railway level crossing system.

Chapter 6: TESTING

6.1 Testing Methodology

Testing is an important stage in the development of the automatic railway level crossing system. It is carried out to verify whether the sensors, microcontroller, and output devices function according to the intended design.

The system is tested by checking the response of the IR sensors, the processing of input signals by the STM32 Blue Pill, and the operation of the servo motor, buzzer, and LEDs. Different operating conditions are considered to verify the sequence of gate closing and opening.

The testing process focuses on the accuracy of train detection, proper activation of warning signals, and correct movement of the railway gate.

6.2 Test Cases

The following test cases are considered for evaluating the operation of the system:

Test Case 1: Normal Condition

- Input: No train is detected.
- Expected Output: Gate remains open, green LED is ON, red LED and buzzer are OFF.

Test Case 2: Train Detection

- Input: First IR sensor detects an approaching train.
- Expected Output: Buzzer and red LED are activated, and the servo motor closes the gate.

Test Case 3: Train Passing Condition

- Input: Train is passing through the crossing.
- Expected Output: Gate remains closed and warning indications remain active.

Test Case 4: Train Clearance Detection

- Input: Second IR sensor detects the train clearance condition.
- Expected Output: Buzzer stops, red LED turns OFF, green LED turns ON, and the servo motor opens the gate.

6.3 Performance Parameters

The performance of the system is evaluated based on the following parameters:

- **Train Detection:** Ability of the IR sensors to detect the approaching train and its clearance.
- **Response Time:** Time taken by the microcontroller to respond to sensor inputs.
- **Gate Movement:** Proper opening and closing of the gate using the servo motor.
- **Warning Indication:** Correct operation of the buzzer and LEDs.
- **System Coordination:** Proper coordination between sensors, the microcontroller, and output devices.

These parameters help evaluate the functionality and reliability of the developed prototype.

Chapter 7: RESULTS AND DISCUSSION

7.1 Functional Testing Results

Functional testing is carried out to evaluate the operation of the Automatic Railway Level Crossing Opening and Closing System using the STM32 microcontroller. The main purpose of testing is to verify whether the system responds correctly to the signals received from the IR sensors and performs the required gate operations.

The system is designed to operate automatically without requiring continuous manual intervention. The two IR sensors act as input devices and provide signals to the STM32 Blue Pill microcontroller. Based on these signals, the microcontroller controls the servo motor, buzzer, and LED indicators.

During normal operation, when no train is detected, the railway gate remains in the open position. The green LED indicates that the crossing is open, while the red LED and buzzer remain inactive. This represents the initial condition of the system.

When the first IR sensor detects an approaching train, the microcontroller processes the sensor input and initiates the gate-closing operation. The buzzer is activated to provide an audio warning, and the red LED turns ON to indicate that the crossing is closed. At the same time, the servo motor moves the gate to the closed position.

The gate remains closed while the train passes through the crossing. After the train passes, the second IR sensor is used to detect the clearance condition. Once the required condition is detected, the microcontroller initiates the gate-opening operation. The servo motor raises the gate, the buzzer is deactivated, the red LED turns OFF, and the green LED turns ON.

These functional conditions are used to evaluate whether the different components work together according to the intended sequence of operation.

7.2 Results

The proposed system demonstrates the basic operation of an automated railway level crossing mechanism. The expected responses under different operating conditions are as follows:

- Normal Condition: The gate remains open, the green LED is ON, and the buzzer and red LED remain OFF.
- Train Detection: The first IR sensor detects the approaching train and sends a signal to the STM32 microcontroller.
- Gate Closing: The buzzer and red LED are activated, and the servo motor moves the gate to the closed position.
- Train Passing: The gate remains closed while the train passes through the crossing.
- Gate Opening: After clearance is detected by the second IR sensor, the gate opens, the buzzer stops, and the green LED turns ON.

These operations demonstrate how sensor inputs and microcontroller-based control can be used to automate railway gate operations.

7.3 Analysis

The system uses IR sensors as input devices and the STM32 Blue Pill as the main processing unit. The microcontroller receives signals from the sensors and controls the output components according to the programmed instructions.

The SG90 servo motor provides controlled movement of the miniature railway gate. The buzzer and LED indicators provide audio and visual warnings to indicate the status of the crossing. The coordination of these components is essential for maintaining the correct sequence of gate operations.

The system demonstrates how automation can reduce dependence on manual gate operation in the prototype. However, proper sensor placement, reliable power supply, and accurate programming are important for achieving the intended operation.

7.4 Key Findings

The following key findings are identified from the design and implementation of the project:

- IR sensors can be used for detecting the approaching train and its clearance.
- The STM32 microcontroller coordinates the operation of multiple hardware components.
- The servo motor enables automatic opening and closing of the miniature gate.
- The buzzer and LEDs provide audio and visual warning indications.
- The system demonstrates the application of embedded systems in railway level crossing automation.
- Automation reduces the need for continuous manual intervention in the prototype.

Overall, the project demonstrates the basic working principle of an automatic railway level crossing system using a microcontroller, sensors, and output devices.

Chapter 8: CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The project titled "Automatic Railway Level Crossing Opening and Closing System Using STM32 Microcontroller" demonstrates the use of embedded systems for automating railway level crossing operations. The system uses IR sensors to detect the approaching train and its clearance, while the STM32 Blue Pill microcontroller processes the sensor inputs and controls the connected output devices.

The SG90 servo motor is used to control the opening and closing movement of the miniature railway gate. A buzzer and LED indicators provide audio and visual warnings to indicate the status of the crossing. The integration of these components demonstrates how hardware and software can work together to perform an automated control operation.

The project provides practical knowledge of microcontroller programming, sensor interfacing, actuator control, and hardware integration. It also demonstrates how automation can reduce dependence on manual intervention in railway gate operation.

Overall, the project successfully demonstrates the basic concept of an automatic railway level crossing system and provides a foundation for developing more advanced automated safety systems.

8.2 Future Scope

The proposed system can be improved further by introducing additional features to increase its reliability, accuracy, and functionality. Some possible future enhancements include:

- **Advanced Train Detection:** More reliable sensors can be used to improve train detection accuracy.
- **Wireless Communication:** Wireless communication can be introduced to share train arrival and crossing status information.
- **Obstacle Detection:** Additional sensors can be installed to detect obstacles on the railway track before the gate is reopened.
- **Emergency Alert System:** An emergency alert mechanism can be included to notify authorities during abnormal conditions.
- **Power Backup:** A backup power supply can be added to maintain system operation during power failures.
- **Real-Time Monitoring:** A monitoring system can be developed to display the current status of the railway crossing.

These improvements can help develop the prototype into a more advanced and reliable railway level crossing automation system. However, practical railway deployment would require extensive safety testing, certified equipment, and approval from the relevant railway authorities.

APPENDICES

APPENDIX A: COMPONENT SUMMARY

The automatic railway level crossing system consists of different hardware components that work together to detect trains and control the railway gate automatically.

Sl. No.	Component	Description
1	STM32 Blue Pill	Main microcontroller used to process sensor inputs and control the system.
2	IR Sensor 1	Detects the approaching train.
3	IR Sensor 2	Detects the train clearance condition.
4	SG90 Servo Motor	Controls the opening and closing movement of the gate.
5	Buzzer	Produces an audio warning when a train approaches.
6	Red LED	Indicates that the railway gate is closed.
7	Green LED	Indicates that the railway gate is open.
8	Resistors	Used to limit current in the LED circuits.
9	Breadboard	Used to establish temporary circuit connections.
10	Jumper Wires	Used to connect the different hardware components.
11	ST-Link V2 Programmer	Used to upload the program to the STM32 microcontroller.
12	Regulated 5V Supply	Provides the required power to the servo motor.

APPENDIX B: SYSTEM FLOW AND ARCHITECTURE

B.1 System Flow

The system follows a sequence of operations based on the signals received from the IR sensors.

1 The system is initialized, and the gate remains in the open position.

2 The first IR sensor detects an approaching train.

3 The STM32 microcontroller processes the sensor signal.

4 The buzzer and red LED are activated.

5 The servo motor moves the gate to the closed position.

6 The gate remains closed while the train passes.

7 The second IR sensor detects the train clearance condition.

8 The microcontroller stops the buzzer and controls the servo motor to open the gate.

9 The red LED turns OFF, and the green LED turns ON.

B.2 System Architecture

The STM32 Blue Pill acts as the central control unit of the system. It receives input signals from the two IR sensors and controls the output devices based on the programmed conditions.

Input Devices:

- IR Sensor 1
- IR Sensor 2

Processing Unit:

- STM32 Blue Pill Microcontroller

Output Devices:

- SG90 Servo Motor
- Buzzer
- Red LED
- Green LED

System Architecture:

IR Sensors → STM32 Blue Pill → Servo Motor, Buzzer and LED Indicators

The architecture represents the connection between the input devices, processing unit, and output devices used in the automatic railway level crossing system.

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